Tutorial Questions | Week 9

COSC2779 - Deep Learning

This tutorial reviews representation learning, foundation models, and how to adapt them (transfer learning, fine-tuning, and parameter-efficient fine-tuning). Please try the questions before you join the session. The question below has a similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a company that builds text-triage tools for a network of community legal centres across Australia. Each centre receives a large volume of free-text client emails, and staff must read every email to decide which legal area it concerns (e.g. *tenancy*, *employment*, *family*, *consumer*, ... – eight areas in total) so it can be routed to the right team.

   You are given:

   - a very large collection of **unlabelled** historical emails (millions), and
   - a small **labelled** set of about **2,000** emails, each tagged with its legal area.

   Constraints: a modest GPU budget, more tasks are planned later (urgency detection, entity extraction), and the setting is high-stakes and fairness-sensitive. You decide to build the system around a pre-trained Transformer.

   (a) Your first instinct is to represent each email by averaging word2vec/GloVe vectors of its words. Why is a contextual representation from a pre-trained Transformer usually better?

   (b) You have millions of unlabelled emails but only 2,000 labels. How can the unlabelled data help? Name a suitable objective.

   (c) What is a **foundation model**, and give two reasons to start from one rather than train a bespoke model from scratch here?

   (d) List the main strategies for adapting a pre-trained model, ordered from cheapest to most expensive, one line each.

   (e) With only ∼2,000 labels and a modest GPU, which strategy (or strategies) would you try, and why?

   (f) Explain the difference between **feature extraction** and **full fine-tuning**, with one pro and one con of each.

   (g) Later you will serve many tasks. Why is full fine-tuning costly to store/serve per task, and how does **LoRA** address this? Write the low-rank update.

   (h) In LoRA, what does the rank r control, and what is the trade-off in increasing it?

   (i) When could **zero-shot** give useful predictions with no training at all, and give one limitation.

   (j) This is a high-stakes, fairness-sensitive setting. Name two risks of using a foundation model here and one mitigation.